

WEEK 2 TECHNICAL EVALUATION

Master Presentation Script & Deep-Dive Q&A; Guide (Banglish)

Presenter: Wasiur Rahman Sakib

Track: 1-Month Web Dev Training

Focus: React 18, Hooks, MongoDB, Next.js

SLIDE 1: Title Slide & Introduction

Badges: [React 18] [Custom Hooks] [MongoDB] [Mongoose] [Next.js] [Vite]

■ ■ Banglish Presentation Speech (■ ■ ■ ■ ■ ■ ■ ■ ■ ■):

"Assalamu Alaikum / Good morning everyone. Ami Wasiur Rahman Sakib, Software Engineer Intern. Ajke ami amar Week 2 Technical Evaluation Presentation share korchir. Ei week-e amader main focus chilo Vanilla JS theke modern component-driven React 18 architecture, Custom Hooks, backend data persistence-er jonno MongoDB & Mongoose, ebong full-stack rendering-er jonno Next.js-e transition kora. Cholen amra step by step core modules, technical challenges ebong practical live projects gulo dekhe nei."

SLIDE 2: Executive Overview (4 Core Modules)

Modules: 1. React Paradigm | 2. State & Hooks | 3. Database Layer | 4. Full-Stack Bridge

■ ■ Banglish Presentation Speech (■ ■ ■ ■ ■ ■ ■ ■ ■ ■):

"Week 2-te amader pura learning ta ke 4 ta main pillar-e divide kora hoyechilo: 1. React Paradigm: Jekhane imperative DOM manipulation theke declarative UI ebong Virtual DOM diffing mechanism bujhechi. 2. State & Hooks: State immutability-r importance, useEffect-er lifecycle sync, ebong logic reuse korar jonno Custom Hooks architecture. 3. Database Layer: NoSQL MongoDB-te BSON document modeling, Mongoose schema validation ebong aggregation pipelines. 4. Full-Stack Bridge: Next.js App Router-er Server vs Client components ebong robust API routes build kora. Ekhon cholen React-er core concepts deep dive kori."

SLIDE 3: React Core & Immutability

Pillars: Uni-directional Flow | Virtual DOM | State Immutability

■ ■ Banglish Presentation Speech (■■■■ ■■ ■■■■):

"React-er predictability ashole 3 ta fundamental concept-er upor depend kore: Prothomoto, Uni-directional Data Flow: Props shob shomoy Parent theke Child-e down hoy, ar Child theke event callback-er maddhome state update trigger hoy. Er karone single source of truth maintain thake ebong data desync bug hoy na. Dwitiyoto, Virtual DOM: React memory-te ekta lightweight virtual UI tree rakhe. State change hole reconciliation algorithm ager tree-r shathe diff check kore, ebong real DOM-e shudhu jei part-e change hoyeche sheita batched way-te update kore. Tritiyoto, State Immutability: JS-e object ba array direct mutate korle reference same theke jay (prev === next), jar fole React re-render skip kore. Tai amra shob shomoy spread operator [...prev] ba pure methods use kori jate notun memory reference toiri hoy ebong time-travel debugging possible hoy."

■ ■■■■■■: Props ■■ ■■■■ Parent ■■■■ Child-■ down ■■■ ■■■■?

Predictability & Single Source of Truth:

- Child
- Parent- (Spaghetti Code)
- Child Read-Only
- Callback

Virtual DOM & Reconciliation

Blue-print Analogy: Real DOM ,  Virtual DOM  -  

■■■■■■: JS-■ Object/Array direct mutate ■■■■ Reference same ■■■■ ■■■■ ■■■■? ■■■■ React State ■■■■ Immutable?

Heap **Pointer** **Mechanism:** Object/Array **Reference** **Type** **Heap** (e.g. 0x0012AF) **React**
user.name = 'Wasiur' **\$O(1)** **Shallow** **Reference** **Comparison** **Spread Operator {...prev}** **Time-Travel Snapshots** **Concurrent Rendering**

SLIDE 4: Lifecycle & Custom Hooks Architecture

Pillars: useEffect Sync | Cleanup Phase | Custom Hooks (useFetch, useToggle)

■■ Banglajish Presentation Speech (■■■■ ■■ ■■■■):

"Side-effects manage korar jonno React-e useEffect use hoy: Amra bujhte perechi je useEffect shudhu lifecycle na, eta ashole External System-er shathe sync korar tool—jemon API call ba browser event listeners. Dependency array control kore amra infinite loop theke bachi. Ekhane shobcheye crucial part holo Cleanup Phase. Component unmount hole ba dependency change hole amra event listener remove kori ebong active fetch request cancel kori, jate kono memory leak na thake. Ebong logic clean ar reusable rakhar jonno amra duita Custom Hook baniechi: useFetch (API data, loading, error state auto manage kore) ebong useToggle (boolean UI state ek line-e handle kore)."

React-Redux: Side-effects?

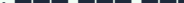
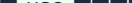
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Cancel

```
useEffect [REDACTED] Cleanup Function return () => { ... } [REDACTED] Event Listener-[REDACTED]
[REDACTED] window.removeEventListener [REDACTED] Fetch Request-[REDACTED] AbortController [REDACTED] active =
false [REDACTED], [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED]
[REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED]
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■ ■ ■ ■ ■: Custom Hook (useFetch ■ useToggle) ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ?

Custom Hook  JS  use 

React Hook  **useToggle:** `const [value, setValue] = useState(init); const toggle = () => setValue(p => !p); return [value, toggle];`  **useFetch:** `data, loading, error`  `useEffect-`  `AbortController` 

 `{ data, loading, error }`

SLIDE 5: MongoDB, Mongoose & Next.js Foundations

Pillars: MongoDB BSON | Mongoose ODM Validation | Next.js Server Components

■■ Banglish Presentation Speech (■■■■ ■■ ■■■■):

"Client-side theke jokhon amra full-stack perspective-e gelam: Amra MongoDB-te BSON documents niye kaj korechi, embedded sub-documents ebong fast query aggregation pipeline bujhechi. Then Mongoose ODM use kore amra data validation layer add korechi—jemon strict schema typing, default values, ebong pre/post save middleware hooks. Ar Next.js-e amra Server Components-er power dekhechi, jekhane heavy logic server-e execute hoy ebong client-e zero unnecessary JavaScript bundle pass hoy, shudhu interactive part gulote client components use kora hoy."

■■■■■■: BSON ■■ ■■■ MongoDB-■■ BSON ■■■■■ ■■■ ■■■ ■■■? Sub-documents ■
 Aggregation Pipeline ■■?

BSON (Binary JSON): JSON-like format supporting Date, ObjectId, Decimal128, Regex
 (Type-Length-Value) Sub-documents:
 Aggregation Pipeline:

■ ■■■■■■: ■■■■ ■■■■■■ Mongoose ODM ■■■■■ Data Validation Layer ■■■■ ■■■■■?

Task.js mongoose Schema Required Custom Error Messages (title), Length Constraints (min 2, max 120), Strict Enum Whitelists (status: 'todo', 'in_progress', 'review', 'done'), Pre-save Middleware (TaskSchema.pre('save', ...))

■ ■ ■ ■ ■: Next.js Server Components vs Client Components (Zero JS Bundle Example)

Server Components ██████████ ██████████ ██████████ ██████████ (await Task.find()) ██████████ ██████████ HTML ██████████, ██████████ ██████████ ██████████ ██████████ JS ██████████ ██████████ ██████████ ██████████ ██████████ ██████████ (onClick ██████████, ██████████ ██████████) ██████████, ██████████ ██████████ ██████████ 'use client' ██████████ ██████████ ██████████ ██████████ (e.g. DeleteTaskButton.jsx) ██████████ ██████████

SLIDE 6: Technical Challenges & Engineering Solutions ■

Core Narrative: Problem Diagnosis ➡ Architecture Solution ➡ Production Code Location

■ ■ Banglish Presentation Speech (■ ■ ■ ■ ■ ■ ■ ■ ■ ■):

"Eibar ashi amader face kora real-world technical blockers ebong amra kivabe egula solve korechi: 1. Challenge 1: Async Race Condition. Problem: User druto click ba parameter change korle ager slow API response pore eshe notun state overwrite kore dito. Solution: useFetch.js-e active cleanup flag ebong AbortController integrate kore pending request instant abort korechi. 2. Challenge 2: Direct State Array Mutation. Problem: Direct array index update korle React re-render trigger korto na ebong move history corrupt hoye jeto. Solution: App.jsx-e spread operator [...squares] ebong history.slice() use kore pure immutable snapshots ensure korechi. 3. Challenge 3: Hot-Reload Connection Leaking in Next.js. Problem: Fast Refresh-e proti save-e notun MongoDB connection toiri hoye pool exhaust hoye jacchilo. Solution: lib/mongodb.js-e global_mongooseClient singleton pattern diye serverless reload-e same connection reuse korechi."

QUESTION: How can we make the most of the information that we have?

- **Race Condition Fix:** [react-playground/src/hooks/useFetch.js](#) (Line 8–32)■
- **State Mutation & Time-Travel Fix:** [react-playground/src/App.jsx](#) (Line 28–33 & 58–62)■
- **DB Connection Pool Leak Fix:** [taskflow-pro/lib/mongodb.js](#) (Line 6–46)■

SLIDE 7: Practical Demonstration (Live Projects & Demo Script)

Live Projects: Tic-Tac-Toe Arena | Custom Hooks Playground | GitHub Pages Host

■ ■ Banglish Presentation Speech (■ ■ ■ ■ ■ ■ ■ ■ ■ ■):

"Ei shob concepts amra live code-e implement korechi: 1. Tic-Tac-Toe Game: Ekhane pure 9-square immutable grid state ache, time-travel move history ache jate user jekono previous move-e jump back korte pare without losing game integrity, ebong 8 winning line check korar pure algorithm ache. 2. Custom Hooks Playground: useFetch ebong useToggle-er dynamic UI demo. Pura project-tai Vite diye bundle kore GitHub Pages-e live deploy kora ache, slide-er bottom link-e click korlei direct live app access kora jabe."

[illegible]

SLIDE 8: Engineering Principles & Best Practices

Principles: 01. Declarative UI | 02. Single Source of Truth | 03. Defensive UI Flow | 04. Modular Logic

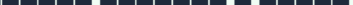
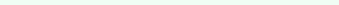
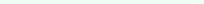
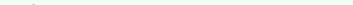
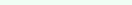
■■ Banglish Presentation Speech (■■■■ ■■ ■■■■):

"Code quality ebong engineering standards maintain korar jonno amra 4 ta core principle follow korechi: 1. Declarative UI: Direct DOM mutate na kore state-driven architecture follow kora. 2. Single Source of Truth: State-ke nearest common parent-e lift-up kora jate sibling component-e data mismatch na hoy. 3. Defensive UI Design: Network error ba slow internet-er jonno shob shomoy loading skeletons ebong fallback error banner ready rakha. 4. Modular Logic Separation: UI view-ke dumb & pure rakhe shob complex side-effects custom hooks-e encapsulate kora."

Engineering Principles

[illegible]

XXXXXXXXXX XXXXXXXXXXXXXXXX:

- **Declarative UI:** App.jsx-
- **Single Source of Truth:** App.jsx-
- **Defensive UI:** CustomHooksDemo.jsx-

- **Modular Logic:** useFetch.js UI

SLIDE 9: Academic Leave & Week 4 Roadmap

Focus: Week 3 Midterms Consolidation ➡ Week 4 React Native Mobile Dev & Full-Stack

■ ■ Banglish Presentation Speech (■ ■ ■ ■ ■ ■ ■ ■ ■ ■):

"Amader upcoming schedule niye bolle: Week 3-te: Ami University Midterm Examination-er jonno approved Academic Leave-e thakbo, shathe exam-er pashe React mental models revise korbo. Week 4-e: Amra full momentum-e back korbo Advanced Next.js Full-Stack & React Native Mobile Development niye—jekhane cross-platform components, API integrations, navigation ebong live cloud backend integration complete kora hobe."

SLIDE 10: Conclusion & Q&A; Transition

Header: THANK YOU! | Questions & Live Project Walkthrough

■ ■ Banglish Presentation Speech (■ ■ ■ ■ ■ ■ ■ ■ ■ ■):

"Dhonnobad shobaikar amar presentation shonar jonno ebong mentors-der continuous guidance-er jonno. Ekhon ami apnader jekono questions ba feedback-er jonno ready achi, ebong chaile live app walkthrough dekhaithe pari. Thank you sir!"